

CStats - Lab #8

April 28, 2026

Exercise 1.

Let the cost function be $f(x) = x^2 - 4x + 4$.

(a) Find the global minimum analytically.

(b) Perform two steps of Gradient Descent by hand. Use an initial starting point of $x_0 = 0$ and the learning rate $\eta = 0.1$.

Exercise 2.

We want to fit a line without a bias term $y = wx$ to a dataset with two data points:

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 4)$$

We use the Mean Squared Error as the loss function for a single data point $L_i(w) = \frac{1}{2}(\hat{y}_i - y_i)^2 = \frac{1}{2}(wx_i - y_i)^2$.

(a) What is the derivative of the loss function with respect to w for a single data point i ?

(b) Start with $w_0 = 0$ and a learning rate $\eta = 0.1$. Perform one update using data point 1, and then a second update using data point 2.

Exercise 3.

We have an input x , one hidden layers, and one output layer such that

$$a_1 = w_1x + b_1$$

$$h_1 = \sigma(a_1)$$

$$\hat{y} = w_2h_1 + b_2$$

The loss function is $L = \frac{1}{2}(\hat{y} - y)^2$ Given a single training example (x, y) , derive the update rules for w_1, b_1, w_2, b_2 using a learning rate η .

Exercise 4.

Suppose you build a Neural Network with 10 hidden layers, but you do not use any activation functions (like ReLU, Sigmoid, Softmax, etc.). Mathematically, what does this 10-layer network simplify to, and why is this a problem for complex datasets?

Exercise 5.

Imagine you are training a model on the large dataset st. 100,000 data points. Explain why calculating the gradient using standard Gradient Descent might be problematic and how does Stochastic Gradient Descent solve this?